

Ordinal Encoding Example

✓ 1 Import necessary libraries

```
1 import pandas as pd
2 from sklearn.preprocessing import OrdinalEncoder
```

✓ 2 Create Sample Data

1. Let's create a DataFrame with educational qualifications,
2. which is a common categorical variable where order matters.
3. We'll use a sample of qualifications in a typical Indian context.
4. The order (10th Pass < 12th Pass < Undergraduate < Postgraduate < Doctorate) is inherent.

```
1 # Sample data representing educational qualifications
2
3 data = {
4     'Student_ID': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
5     'Qualification': [
6         '10th Pass', 'Undergraduate', 'Postgraduate', '12th Pass', 'Doctorate',
7         'Undergraduate', '10th Pass', 'Postgraduate', '12th Pass', 'Undergraduate'
8     ]
9 }
10
11 df = pd.DataFrame(data)
12 display(df)
```

	Student_ID	Qualification
0	101	10th Pass
1	102	Undergraduate
2	103	Postgraduate
3	104	12th Pass
4	105	Doctorate
5	106	Undergraduate
6	107	10th Pass
7	108	Postgraduate
8	109	12th Pass
9	110	Undergraduate

3 Define the Ordinal Order

1. When using `OrdinalEncoder`, it's crucial to explicitly define the order of categories if there's a natural ranking.
2. This ensures that the encoding is done correctly according to the inherent order.
3. Here, we define the order for 'Qualification' from lowest to highest educational level.

```

1 # Define the specific order for the 'Qualification' column
2
3 qualification_order = [
4     '10th Pass',
5     '12th Pass',
6     'Undergraduate',
7     'Postgraduate',
8     'Doctorate'
9 ]
10

```

- Initialize the `OrdinalEncoder` with the specified categories order
- We pass a list of lists because `OrdinalEncoder` expects a list of categories for each feature

```

1 encoder = OrdinalEncoder(
2     categories=[qualification_order]

```

4 Fit and transform the 'Qualification' column

```
1 df['Qualification_Encoded'] = encoder.fit_transform(df[['Qualification']])
```

```
1 display(df)
```

	Student_ID	Qualification	Qualification_Encoded
0	101	10th Pass	0.0
1	102	Undergraduate	2.0
2	103	Postgraduate	3.0
3	104	12th Pass	1.0
4	105	Doctorate	4.0
5	106	Undergraduate	2.0
6	107	10th Pass	0.0
7	108	Postgraduate	3.0
8	109	12th Pass	1.0
9	110	Undergraduate	2.0

Explanation

- **Qualification_Encoded**: This new column contains the numerical representation of the 'Qualification' column based on the order we defined:
 - 10th Pass is encoded as 0.0
 - 12th Pass is encoded as 1.0
 - Undergraduate is encoded as 2.0
 - Postgraduate is encoded as 3.0
 - Doctorate is encoded as 4.0

This method is suitable when the categories have a clear, meaningful order, and you want to preserve that order in the numerical representation for machine learning models.

1 Start coding or generate with AI.